

SAT Unit 1.2

Linear Function

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Overview

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- 2 Slope
- 3 Graphs of Linear Function
- 4 Practice

Linear Function

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Linear Functions

- A linear function is an equation that models a **straight line**.
- General form:

$$f(x) = mx + b$$

where m is the slope and b is the y-intercept.

- Properties:
 - Constant rate of change.
 - Graph is a straight line.
 - Slope m determines the steepness.

Evaluating Functions

Topic: Evaluating a Function at a Given Input

- A function such as $h(x) = 4x - 9$ describes a relationship between input x and output $h(x)$.
- To evaluate the function at a specific value (e.g., $x = -3$), substitute that value into the expression.
- Carefully apply arithmetic operations: multiplication first, then addition/subtraction.
- This process gives the function's output for that input, also known as the value of the function.

Key Skill: Mastery of function evaluation builds the foundation for function tables, graphing, and understanding function behavior.

Question : X-Intercept of Combined Functions

The functions f and g are defined as:

$$f(x) = \frac{1}{4}x - 9, \quad g(x) = \frac{3}{4}x + 21.$$

If the function h is defined as:

$$h(x) = f(x) + g(x),$$

what is the x-coordinate of the x-intercept of the graph of $y = h(x)$ in the xy -plane?

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Understanding Slope

Definition: The **slope** of a line measures the steepness and direction of the line. It is represented as:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

where:

- m is the slope,
- (x_1, y_1) and (x_2, y_2) are two points on the line,
- Δy represents the vertical change (rise),
- Δx represents the horizontal change (run).

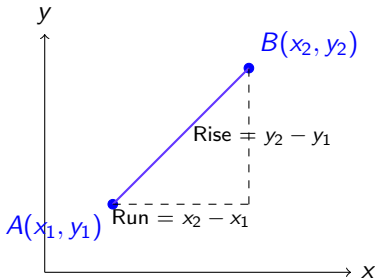
Real-World Applications of Slope:

- **Road Inclines:** The slope of a highway ramp or a hill indicates how steep the road is.
- **Construction:** Roofs have slopes to allow rainwater to flow off properly.

Understanding Slope Visually

Slope (m) measures the steepness and direction of a line.

- Formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Also called: **rise over run**
- Positive slope: line rises from left to right
- Negative slope: line falls from left to right



Slope shows how much y increases or decreases as x increases.

Question : Temperature Conversion

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins:

$$F(x) = \frac{9}{5}(x - 273.15) + 32.$$

If a temperature increased by **2.1** kelvins, by how much did the temperature increase, in degrees Fahrenheit?

- A.** 3.78
- B.** 35.78
- C.** 487.89
- D.** 519.89

Question: Meaning of the Constant in the Equation

An object hangs from a spring. The formula

$$\ell = 30 + 2w$$

relates the length ℓ , in centimeters, of the spring to the weight w , in newtons, of the object.

Which of the following describes the meaning of the 2 in this context?

Options:

- A. The length, in centimeters, of the spring with no weight attached.
- B. The weight, in newtons, of an object that will stretch the spring 30 centimeters.
- C. The increase in the weight, in newtons, of the object for each one-centimeter increase in the length of the spring.
- D. The increase in the length, in centimeters, of the spring for each one-newton increase in the weight of the object.

Answer Explanation: Meaning of the Constant in the Equation

Correct Answer: D

Solution:

- The given equation is $\ell = 30 + 2w$.
- The coefficient 2 represents the rate at which the length of the spring increases as the weight increases.
- This means that for each additional newton of weight, the length of the spring increases by 2 centimeters.

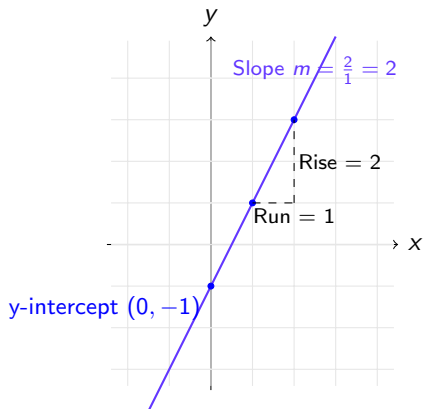
Final Answer: D.

Graphs and Applications of Linear Function

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Graphing a Linear Function

Example: Graph of $f(x) = 2x - 1$



A linear function graphs as a straight line. Slope and intercept define its position.

Interpreting Piecewise Linear Graphs

Topic: Interpreting Piecewise Linear Graphs & Calculating Rate of Change

- A graph showing a quantity over time helps visualize how that quantity changes.
- The **rate of change** (e.g., rainfall per hour) is calculated as:

$$\text{Rate} = \frac{\text{Change in } y}{\text{Change in } x} = \frac{\Delta y}{\Delta x}$$

- A **horizontal segment** means the quantity is not changing during that time (rate = 0).
- A **steeper line** means a faster rate of change.
- When analyzing the graph:
 - Look for intervals with different slopes.
 - Compare slope values to find where the rate is greatest or increasing/decreasing.

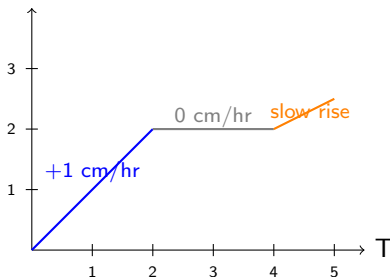
Key Skill: Understanding graphs requires both visual reading and numerical slope calculation.

Interpreting Piecewise Linear Graphs

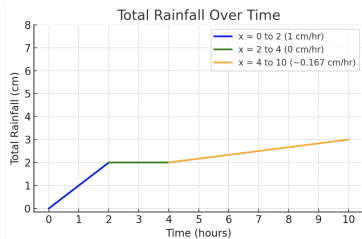
Concept: Use graphs to visualize how quantities change over time.

- A line segment shows a constant rate of change.
- A steeper slope means a faster rate.
- A flat segment means the value is not changing (rate = 0).
- Rate = $\frac{\Delta y}{\Delta x}$

Total Rainfall (cm)



Question : Rainfall Rate Interpretation



The graph shows the total amount of rainfall y , in centimeters, from the start of a 10-hour period, where x is the number of hours after the start of the period. Which of the following statements about the rainfall during this time period is true?

- A. The rate of rainfall was 2 centimeters per hour between $x = 2$ and $x = 4$.
- B. The rate of rainfall was the greatest between $x = 4$ and $x = 10$.
- C. The rate of rainfall increased between $x = 0$ and $x = 2$.
- D. The rate of rainfall was 0 centimeters per hour between $x = 2$ and $x = 4$.

Answer : Rainfall Rate Interpretation

Analyzing the Graph:

- From $x = 0$ to $x = 2$:

$$\text{Rate} = \frac{2 - 0}{2 - 0} = 1 \text{ cm/hr}$$

- From $x = 2$ to $x = 4$:

$$\text{Rate} = \frac{2 - 2}{4 - 2} = 0 \text{ cm/hr (flat segment)}$$

- From $x = 4$ to $x = 10$:

$$\text{Rate} = \frac{3 - 2}{10 - 4} = \frac{1}{6} \approx 0.167 \text{ cm/hr}$$

Evaluating the Choices:

- A.** Incorrect — rate was 0, not 2
- B.** Incorrect — greatest rate was from $x = 0$ to 2
- C.** Incorrect — rate did not increase; it was constant at 1
- D. Correct** — rate was 0 cm/hr between $x = 2$ and $x = 4$

Question : Window Repair Cost

A window repair specialist charges **\$220** for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is **\$400**. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

- A. $f(x) = 60x + 100$
- B. $f(x) = 60x + 220$
- C. $f(x) = 80x$
- D. $f(x) = 80x + 220$

Answer to Question

To determine the hourly fee, subtract the base cost from the total cost for 5 hours:

$$400 - 220 = 180 \quad \Rightarrow \quad \text{for 3 extra hours}$$

$$\text{Hourly fee} = \frac{180}{3} = 60$$

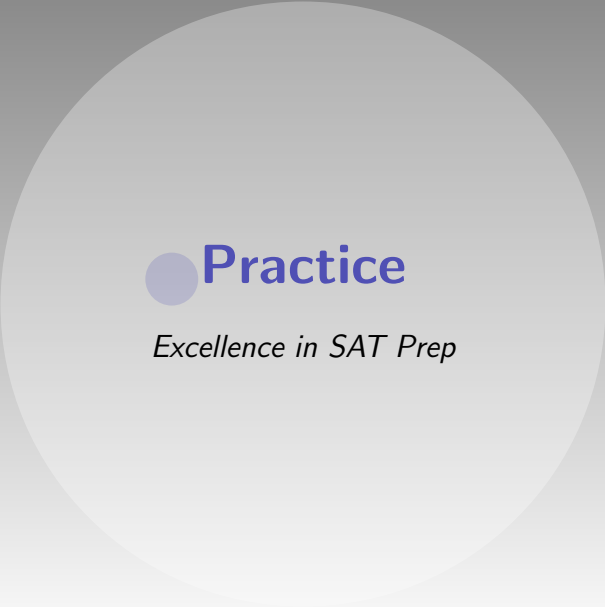
So, the cost function is:

$$f(x) = 220 + 60(x - 2)$$

Simplifying:

$$f(x) = 60x + 100$$

Correct Answer: A. $f(x) = 60x + 100$



Practice

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Question: Interpreting Constants in a Function

The given function $h(t) = 100 - 4t$ models the number of hours until a gallon of milk held at a temperature of t , in degrees Celsius, goes sour.

Which of the following is the best interpretation of the number 4 in this context?

- A. An increase of 1°C will make a gallon of milk go sour 4 hours faster.
- B. An increase of 1°C will make 4 gallons of milk go sour 1 hour faster.
- C. An increase of 4°C will make a gallon of milk go sour 1 hour faster.
- D. An increase of 4°C will make a gallon of milk go sour 4 hours faster.

Answer: Interpreting Constants in a Function

Correct Answer: **A**

Reasoning:

- The function is $h(t) = 100 - 4t$, which means the number of hours until souring *decreases* as temperature increases.
- The coefficient -4 represents the rate of change: for each 1°C increase in temperature, the milk sours 4 hours **sooner**.
- This matches exactly with Choice A.

Interpretation: The number 4 means that **for every 1°C rise in temperature, the souring time decreases by 4 hours.**

Question : Phone Plan Cost Increase

The function $C(x) = 0.07(x - 180) + 62.6$ gives the monthly cost, in dollars, of using an international phone plan for x minutes each month.

If the phone plan is used for 75 more minutes each month, by how many dollars does the monthly cost of the plan increase?

- A. 1.75
- B. 5.25
- C. 55.25
- D. 67.85

Answer

Correct Answer: B.

Reasoning:

- The slope of the function is 0.07, which is the cost per additional minute.
- An increase of 75 minutes leads to:

$$\Delta C = 0.07 \times 75 = 5.25$$

- So the cost increases by \$5.25.

Question : Shipment Cost and Weight

The function $C(x) = 1.5 + 2.5x$ models the cost, in dollars, of mailing a shipment weighing x pounds.

According to the model, an increase of 10 dollars in the mailing cost corresponds to an increase of how many pounds in the weight of the shipment?

- A. 2
- B. 2.5
- C. 4
- D. 5

Answer

Correct Answer: D.

Reasoning:

- The slope 2.5 is the cost per pound.
- To find the weight change that corresponds to a \$10 increase:

$$\frac{10}{2.5} = 4 \quad (\text{Oops!})$$

Correction:

$$\frac{10}{2.5} = \boxed{4}$$

Correct Answer is: C. 4, not D.

Question: Alternating Context – Lawn Care Service

A lawn care company charges **\$150** for the first two hours of work and an additional hourly rate for each hour after that. The total charge for 6 hours of work is **\$390**. Which function f gives the total cost, in dollars, for x hours of work, where $x \geq 2$?

- A. $f(x) = 80x + 150$
- B. $f(x) = 80x + 30$
- C. $f(x) = 60x + 30$
- D. $f(x) = 60x + 150$

Answer: Lawn Care Service Question

The total cost for 6 hours is \$390, and the first 2 hours cost \$150. So, the cost for the additional 4 hours is:

$$390 - 150 = 240$$

$$\text{Hourly rate} = \frac{240}{4} = 60$$

The cost function is:

$$f(x) = 150 + 60(x - 2)$$

Simplifying:

$$f(x) = 60x + 30$$

Correct Answer: C. $f(x) = 60x + 30$

Question: Alternating Context – Aquarium Fish Population

A marine biologist observes an initial population of **900 fish** in a large aquarium. The biologist's goal is to increase the population to **2,100 fish**. The population model assumes:

- the population begins at 900
- it increases by 150 fish per week for the first 3 weeks
- after that, it increases by 200 fish per week until the goal is reached

According to this model, at the end of week w after the initial observation, where $w > 3$, which of the following functions gives the predicted number of fish still needed to reach the biologist's goal?

- A. $p(w) = 1200 - 200w$
- B. $p(w) = -150 + 200w$
- C. $p(w) = 1350 - 200w$
- D. $p(w) = 2100 - 200w$

Answer: Aquarium Fish Population

Step 1: Calculate fish after the first 3 weeks

Initial population: 900

Increase over 3 weeks: $3 \times 150 = 450$

Population after 3 weeks: $900 + 450 = 1350$

Step 2: Fish still needed after week 3:

$$2100 - 1350 = 750$$

Step 3: Model for weeks beyond 3 Each week adds 200 more fish. So fish still needed is:

$$p(w) = 750 - 200(w - 3) = 750 - 200w + 600 = 1350 - 200w$$

Correct Answer: C. $p(w) = 1350 - 200w$

Summary of Unit 1.2: Linear Functions

- **Linear functions** model constant rates of change — the graph is always a straight line.
- The general form $f(x) = mx + b$ helps identify slope (m) and y-intercept (b).
- **Slope** $m = \frac{\Delta y}{\Delta x}$: Describes steepness and direction of a line.
- Real-world problems often involve interpreting slope, intercept, or evaluating the function.
- Graph interpretation includes understanding intervals of change and comparing rates.
- Applications range from temperature conversion to business pricing models and population growth.

Thank You!

We appreciate your time and focus.

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